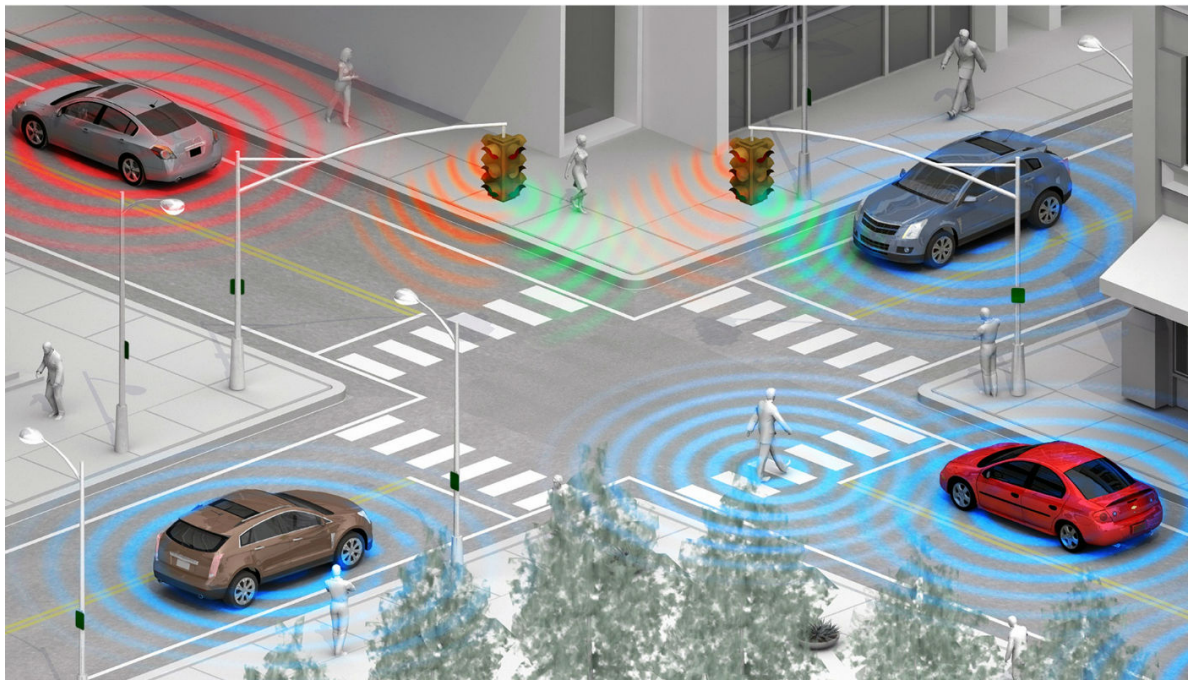
Internet of Things and Vehicular Networks (TP-546)

Samuel Baraldi Mafra

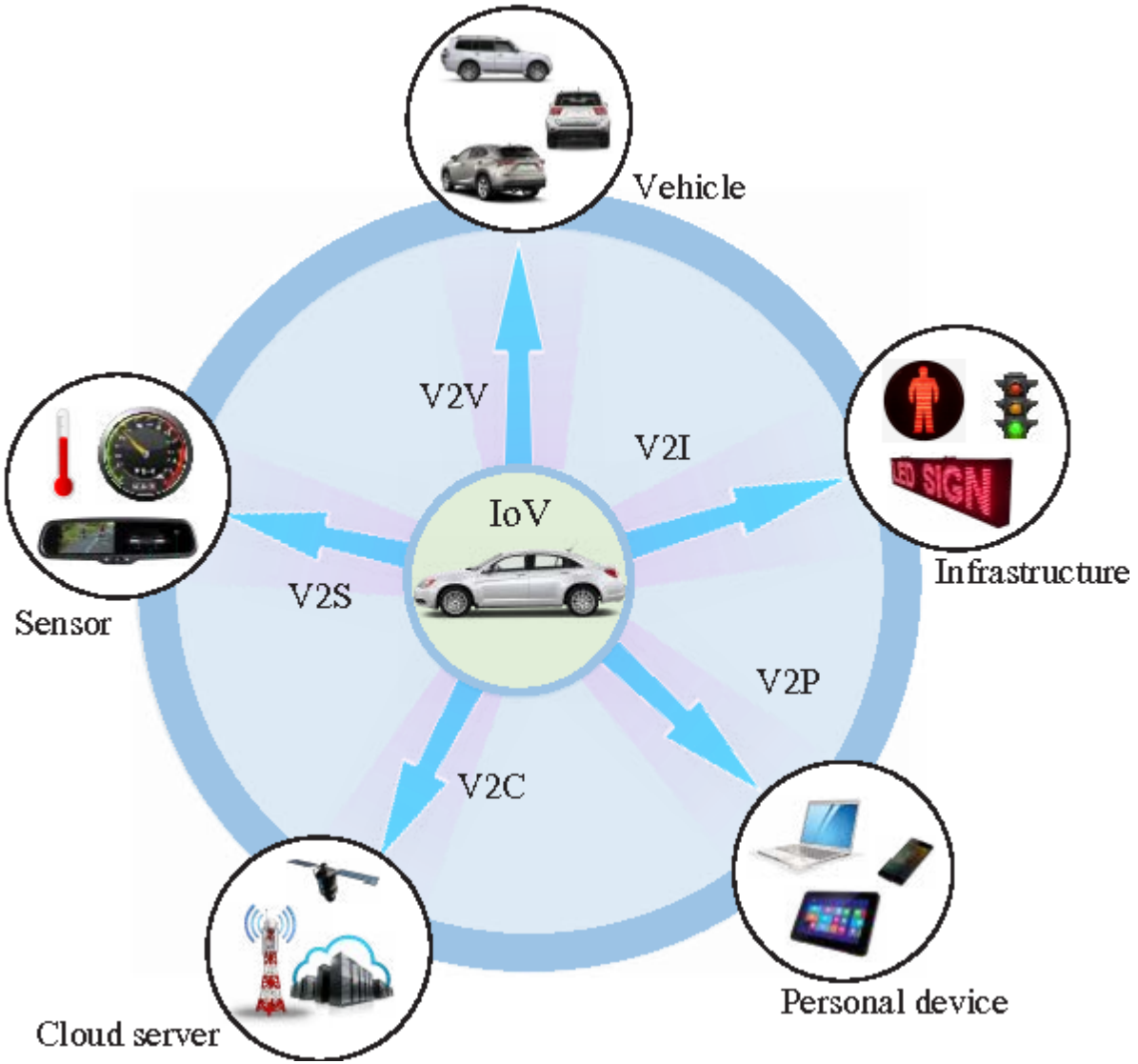
Inatel
Instituto Nacional de Telecomunicações

Vehicular Networks

Vehicular networks are networks formed by motor vehicles and fixed equipment generally located alongside streets or roads.



Internet of vehicles (IOV)



Opportunistic networks or delay-tolerant networks.



Opportunistic networks:

- An opportunistic network is a network of wirelessly connected nodes.
- The communication range between two connected nodes extends only over a short distance.
- Nodes are connected only temporarily, and the network topology may change because of node mobility or node activation and deactivation.

- Node discovery: A network node can discover other network nodes within direct communication range.
- Single-hop message exchange: A node can send and receive arbitrary data to or from any other node within direct communication range.
- Multihop communication.

Delay-tolerant networks:

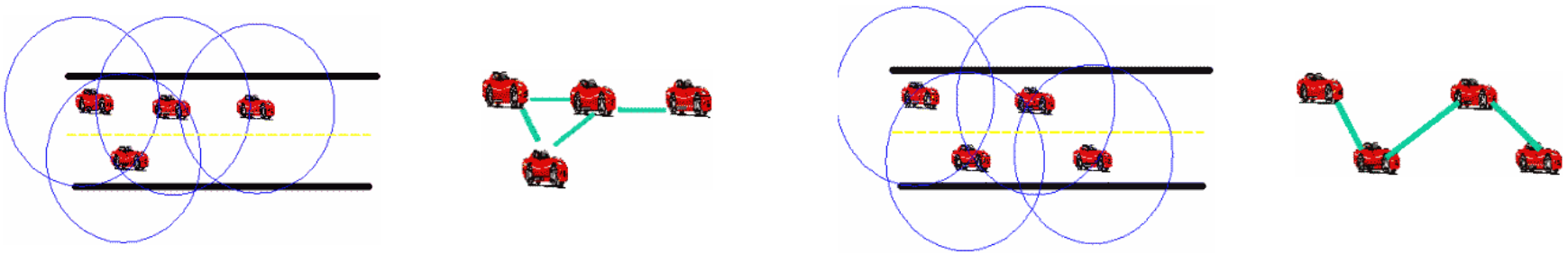
- Delay-tolerant networking (DTN) is a computer network architecture that addresses technical problems in heterogeneous networks where continuous network connectivity may be unavailable (Internet outages).

Factors to consider when categorizing a network as delay-tolerant

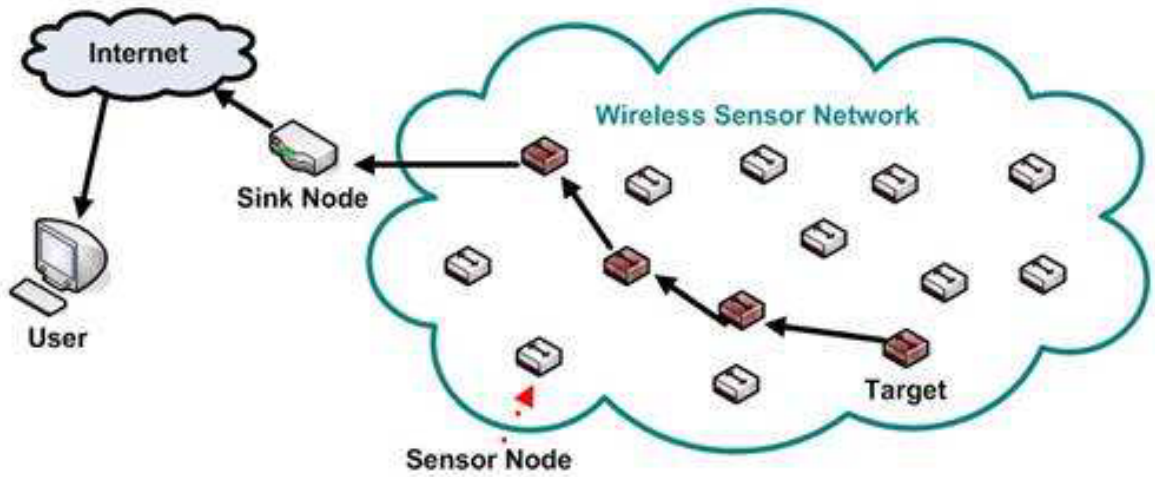
- Age of information is defined as the time elapsed since the latest update packet successfully received at the monitor was generated at the source;
- Importance of the message to be sent.

Examples of opportunistic networks

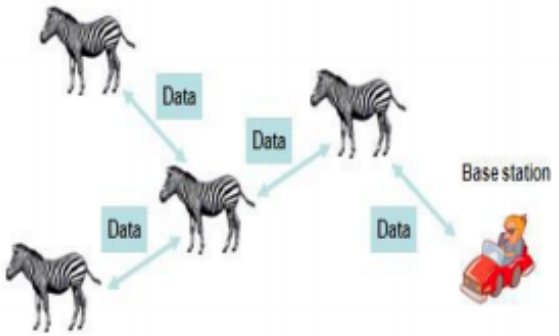
- Rapid topology changes. High relative vehicle speeds lead to short link lifetimes;
- Frequent fragmentation. Parts of the network cannot reach nodes in nearby regions.
- Small effective network diameter. A path may cease to exist almost as quickly as it was discovered (reactive routing).



Wireless sensor networks



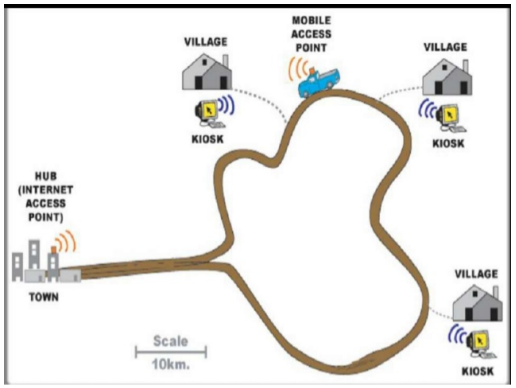
Environmental monitoring: ZebraNet





- Special GPS-equipped collars were fitted to zebras.
- 35,000 zebras over approximately 40,000 square kilometers.

DakNet



- DakNet, an ad hoc network that uses wireless technology to provide asynchronous digital connectivity, demonstrates that combining wireless and asynchronous services can indeed be the beginning of a path toward universal broadband connectivity.

- DakNet has been successfully deployed in remote parts of India and Cambodia at a cost two orders of magnitude lower than traditional fixed-line solutions.

Satellite communication



Wars/disasters





Factors that contribute to difficulties in message dissemination:

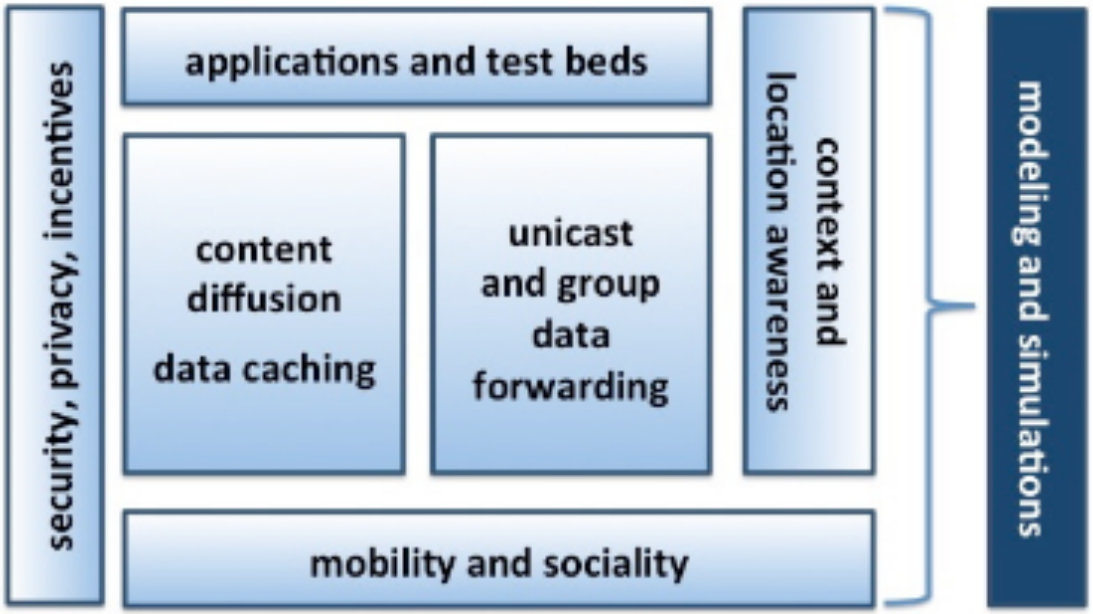
- Node mobility;
- Running out of battery power;
- Inherent instability of wireless links.

End-to-end path connectivity between the message source and destination may never exist.

- Opportunistic networks turn this obstacle, namely mobility, into an opportunity by allowing nodes to carry messages with them as they move, until a suitable next hop is found.

- Store, carry, and forward paradigm.

Research opportunities and challenges



Mobility and sociality:

- Connectivity is supported by communication technologies that allow direct communication between devices, with Bluetooth and WiFi being the most popular solutions.
- Communication channels are established when two nodes encounter each other, that is, when they enter each other's radio range.
- Because nodes are sparsely distributed throughout the system, encounters are sporadic and depend on the nodes' mobility patterns.

- The importance of mobility in opportunistic networks is twofold.
- On the one hand, understanding mobility is essential to identify what may affect the performance of opportunistic data delivery schemes.
- For example, the time elapsed between consecutive encounters for a given pair of nodes plays a crucial role.
- These time intervals are the main component of the delay accumulated by messages. Identifying their characteristics from real mobility traces is key to investigating the environment in which data delivery protocols operate.

- Understanding mobility is also essential for evaluating the performance of protocols for opportunistic networks under realistic conditions.
- To this end, mobility characteristics must be incorporated into a mobility model to increase the accuracy of simulators.

Communication.

- The solutions proposed in the literature can be classified according to two main paradigms, namely:
  - People-centric;
  - Data-centric.

- The people-centric approach is typical of data forwarding ("User A wants to communicate with user B"), in which the two communication endpoints explicitly address each other's identities.

- Group communications (for example, similar to anycast or multicast) are also considered in some proposals.

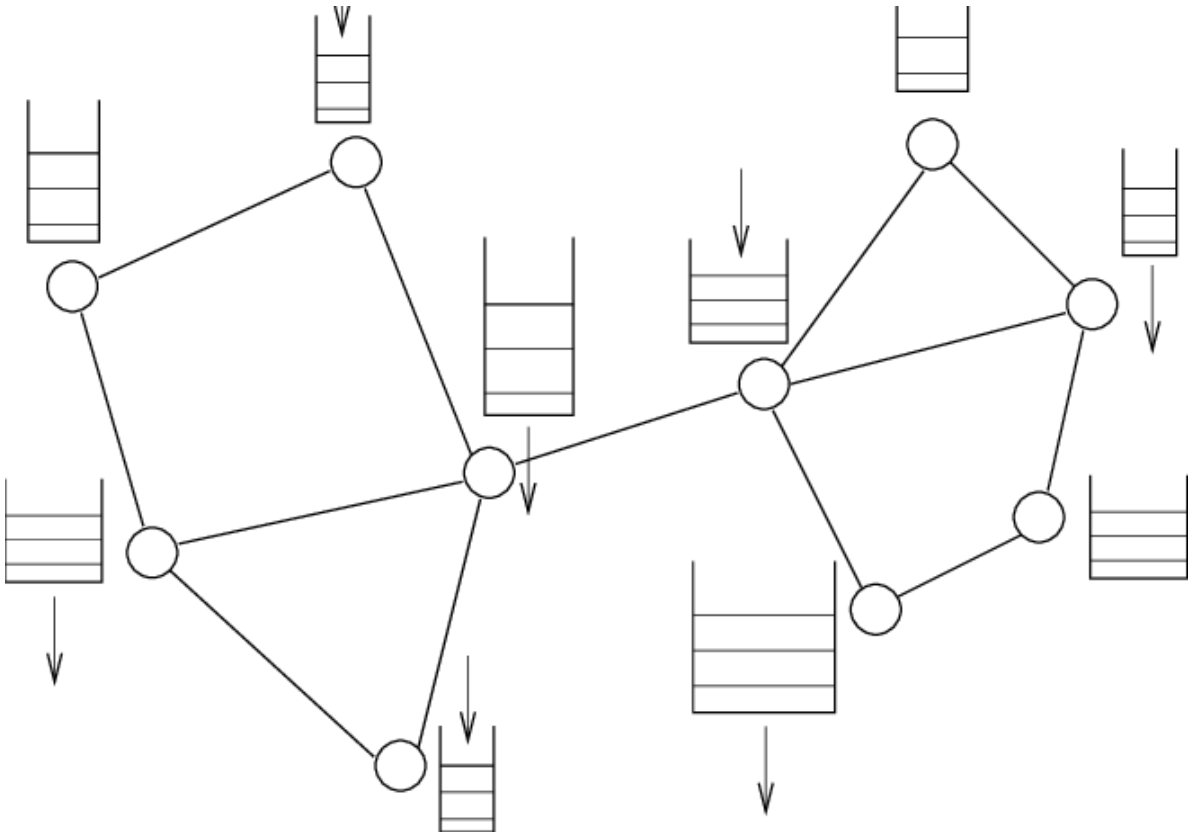
- In the data-centric approach (which can be framed within the popular idea of Content-Centric Networking), what matters is not the identities of the endpoints, but the semantics of the content itself.

- Thus, content is rapidly produced by users and requested by other users based on their interest in those data. This is the content dissemination perspective.

Example: Disseminating warning messages in vehicular networks.

- Both families of data exchange policies can be affected by the mechanisms adopted to manage limited message buffers and cache replacement.

- The importance of buffer planning.
- Buffers are as important in opportunistic networks as batteries are in wireless networks.



The main objectives of buffer management are:

- (a) Remove redundant information from the system;
- (b) Formulate a reasonable queueing strategy;
- (c) Control congestion;
- (d) Develop a cache replacement policy.

- Data forwarding schemes can be classified into two broad categories: socially unaware and socially aware schemes (also known as random schemes versus service-based schemes), based on the amount of information they exploit to make forwarding decisions.

- In socially unaware protocols, a certain number of message replicas (possibly unlimited) are randomly disseminated throughout the environment, in the hope that one of the replica holders will reach the destination.

- The performance of these solutions generally improves as the number of replicas increases.

- In contrast, in socially aware protocols, nodes try to estimate the different probabilities that other nodes will encounter the destination of a buffered message, so as to relay it to the most promising next hop in terms of the probability of successful delivery.

- The estimate is generally inferred from observations of previous encounter patterns, assuming that people exhibit movement habits that repeat in some way over time.

- Socially aware protocols are typically more accurate in reaching the destination, but at the cost of storing a potentially large amount of contextual information.

Context and location awareness.

- Context learning and management involve acquiring and updating a large amount of information about the environment in which users are immersed. Such information may include, for example, how frequently other nodes are encountered, the strength of social relationships between users, regularly visited locations, the type of content encountered nodes are interested in, etc. This wealth of information underpins some of the most interesting operating modes of opportunistic networks.

Location

- The ability to determine a device's current position can be used to filter data available on other users' devices and environmental devices, in order to customize the services and content provided to the user by the opportunistic network.

- Analyzing this aspect is useful for designing applications and data dissemination/forwarding mechanisms.

Security and privacy

- Security in opportunistic networks encompasses privacy, confidentiality, and the enforcement of trust and cooperation. In opportunistic networks, nodes make forwarding decisions based on packet content and/or context.

- Confidentiality and privacy requirements mean that such information must be encrypted. Therefore, the major challenge is to resolve the conflict between encryption and opportunistic forwarding, allowing network mechanisms to operate on encrypted data without decrypting them.

- Another critical aspect of opportunistic networks is the lack of infrastructure. Nodes form a temporary network without the help of any security infrastructure and without any prior business relationship.

- The lack of security infrastructure and the scarcity of resources would inherently encourage selfish behavior among nodes.

- Encouraging cooperation among nodes can rely on reward and/or reputation mechanisms and on the use of additional contextual information, such as the social contacts (friends) of the communicating nodes.

Routing protocols:

- The delay-tolerant networking architecture defines several types of contacts: scheduled, opportunistic, and predicted.
- Scheduled contacts may exist, for example, between a base station somewhere on Earth and a low-Earth-orbit relay satellite.
- Opportunistic contacts arise simply from the presence of two entities in the same place, in an encounter that was neither scheduled nor predicted.
- Finally, predicted contacts are also unscheduled, but their occurrence can be predicted by analyzing previous observations.

- Because the network suffers from connectivity problems, routing algorithms such as the Optimized Link State Routing Protocol (OLSR), based on disseminating control information, or Ad hoc On-Demand Distance Vector (AODV), which operates on demand, fail to achieve routing.

- In flooding-based schemes, a node receives packets, stores them in its buffers, carries them as it moves, and forwards them to other nodes when they meet.

- Excessively spreading multiple copies throughout the network causes severe congestion and exhausts nodes' buffer space, dramatically affecting transmission performance.

- Opportunistic: A node waits to encounter the destination in order to transfer its packet. The main advantage of this method is that it involves only one transmission per packet.

- Random: When a node is at a location and the packet destination is not there, the node transfers the packet to a randomly selected neighbor. A rule prevents local loops: a node can handle a packet only once per visit to the location.

- MobySpace[1];
- Disseminate information to nodes with mobility patterns similar to those of the destination node;
- Euclidean space;
- Nodes are mobile and have wireless networking capabilities.
- They can communicate with each other only when they are within transmission range.
- The network experiences frequent connectivity disruptions, making its topology intermittent and partially connected.

[1] J. Leguay, T. Friedman and V. Conan, "Evaluating Mobility Pattern Space Routing for DTNs," Proceedings IEEE INFOCOM 2006. 25TH IEEE International Conference on Computer Communications, Barcelona, 2006, pp. 1-10, doi: 10.1109/INFOCOM.2006.299.

- If we want to route a packet from one node to another, it is useful to view the information about these contacts as forming a virtual space.
- Each possible contact is an axis, and the distance along that axis indicates a measure of contact probability.
- Two nodes with similar sets of contacts, encountered at similar frequencies, are close in this space. Nodes with very different contact sets, or that encounter the same contacts at very different frequencies, are far apart.
- Knowledge of node movement along different axes.

- If nodes' visits to specific locations can be tracked, a node's mobility pattern can be described by its visits to those locations.
- In this scenario, each axis represents a location, and the distance along the axis represents the probability of finding a node at that location.
- We can imagine that nodes with similar probabilities of visiting similar sets of locations are more likely to encounter one another than nodes that differ greatly in these respects.

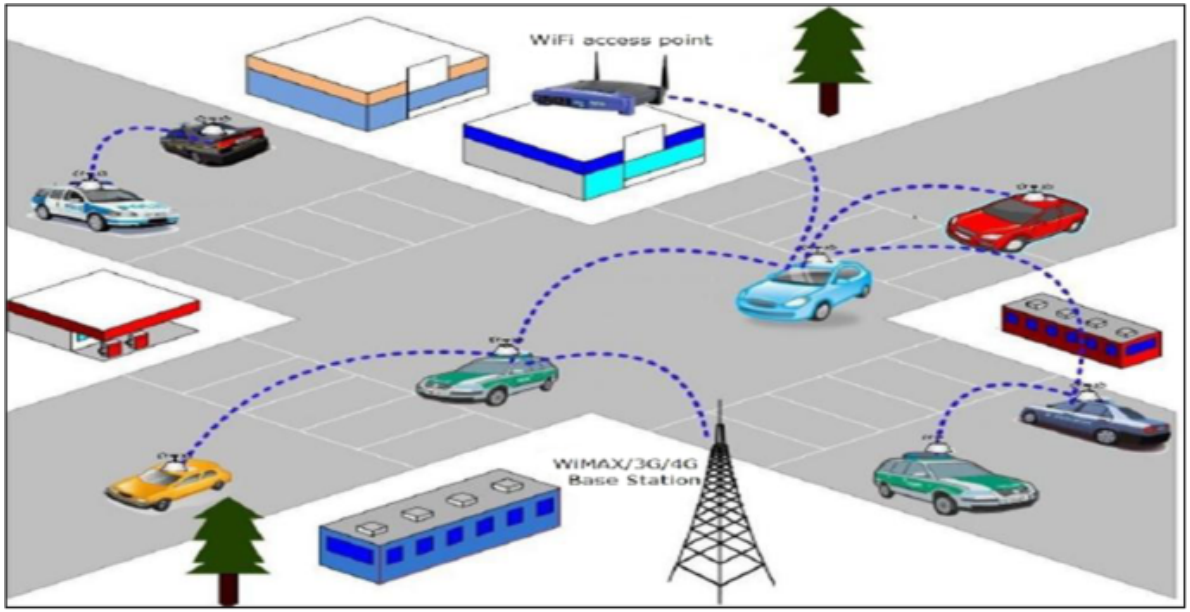
- Shortest Expected Path Routing (SEPR) [2]
- Nodes with the shortest expected path length to the destination are selected as forwarders.
- Location and probability concepts;

[2] Kun Tan, Qian Zhang, & Wenwu Zhu. (n.d.). Shortest path routing in partially connected ad hoc networks. GLOBECOM 03. IEEE Global Telecommunications Conference (IEEE Cat. No.03CH37489).

- MaxProp[3];
- Estimate neighbors' highest probability of delivery to the destination based on the frequency of encounters with the destination;
- MaxProp relies on prioritizing both the scheduling of packets transmitted to other peers and the scheduling of packets to be dropped.
- These priorities are based on path probabilities to peers derived from historical data, as well as several complementary mechanisms, including acknowledgments, a head start for new packets, and lists of previous intermediaries.

[3]Burgess, J.; Gallagher, B.; Jensen, D.; Levine, B.N. Maxprop: Routing for vehicle-based disruption-tolerant networks. In Proceedings of the 25th IEEE International Conference on Computer Communications, Barcelona, Spain, 23?29 April 2006; pp. 1?11.

- Vehicular network routing protocols
  - Location;
  - Topology;
  - Infrastructure;
  - Vehicle density.


WiFi access point
WiMAX/3G/4G
Base Station

- GeoSpray[5]
- Makes routing decisions based on geographic location data and combines multiple-copy and single-copy schemes in a hybrid approach.
- First, it starts with a multiple-copy scheme, spreading a limited number of packet copies to explore alternative paths.
- It then switches to a forwarding scheme that takes advantage of additional contact opportunities.

[5]Soares, V. N. G. J., Rodrigues, J. J. P. C., & Farahmand, F. (2014). GeoSpray: A geographic routing protocol for vehicular delay-tolerant networks. Information Fusion, 15, 102?113. doi:10.1016/j.inffus.2011.11.003

GeoSpray is based on the following design principles:

- Support an opportunistic networking paradigm (in which vehicles are used opportunistically to carry data between nodes) and packet delivery based on the store, carry, and forward paradigm.
- Use geographic location information provided by positioning devices to make routing decisions.
- Employ a multiple-copy routing scheme, with a strict upper limit on the number of copies per packet, combined with a forwarding strategy to improve packet delivery time over multihop routes.
- Delete packets (at intermediate nodes) that have already been delivered to their destinations.